

Intolerance of cow's milk and chronic constipation in children.

BACKGROUND: Chronic diarrhea is the most common gastrointestinal symptom of intolerance of cow's milk among children. On the basis of a prior open study, we hypothesized that intolerance of cow's milk can also cause severe perianal lesions with pain on defecation and consequent constipation in young children. **METHODS:** We performed a double-blind, crossover study comparing cow's milk with soy milk in 65 children (age range, 11 to 72 months) with chronic constipation (defined as having one bowel movement every 3 to 15 days). All had been referred to a pediatric gastroenterology clinic and had previously been treated with laxatives without success; 49 had anal fissures and perianal erythema or edema. After 15 days of observation, the patients received cow's milk or soy milk for two weeks. After a one-week washout period, the feedings were reversed. A response was defined as eight or more bowel movements during a treatment period. **RESULTS:** Forty-four of the 65 children (68 percent) had a response while receiving soy milk. Anal fissures and pain with defecation resolved. None of the children who received cow's milk had a response. In all 44 children with a response, the response was confirmed with a double-blind challenge with cow's milk. Children with a response had a higher frequency of coexistent rhinitis, dermatitis, or bronchospasm than those with no response (11 of 44 children vs. 1 of 21, $P=0.05$); they were also more likely to have anal fissures and erythema or edema at base line (40 of 44 vs. 9 of 21, $P<0.001$), evidence of inflammation of the rectal mucosa on biopsy (26 of 44 vs. 5 of 21, $P=0.008$), and signs of hypersensitivity, such as specific IgE antibodies to cow's-milk antigens (31 of 44 vs. 4 of 21, $P<0.001$). **CONCLUSIONS:** In young children, chronic constipation can be a manifestation of intolerance of cow's milk.